Cheap Metrics First, Replanning Diagnostics Second: Offline Checkpoint Screening for Robot Imitation Learning

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***Abstract*—We study how to screen saved robot-policy checkpoints for rollout evaluation using held-out demonstrations. We compare action error, temporal regularity on recorded observations, and Policy-in-the-Loop Latent Validation (PLV), which feeds policy actions through learned dynamics and returns predicted observations to the policy. Across 1,365 low-dimensional robomimic checkpoints from 45 training runs, with 50 simulator rollouts per checkpoint, action error and open-loop smoothness are strong baselines. Modeled-feedback smoothness improves success-rank correlation on Diffusion Policy (0.556 versus 0.462 for absolute action error), while cheaper metrics lead on recurrent behavior cloning (BC-RNN). Pooled retention at a 30% rollout budget instead favors open-loop smoothness. For action-chunking policies, we also compare plans before and after an eight-action commitment. Boundary jump and replan overlap track success in a hard-task discovery pool, but matched expert, model, and simulator feedback controls do not reproduce that ordering across cohorts. In a 47-checkpoint same-start audit, doubling dynamics-model width reduces 16-step state-prediction MSE from 3.135 to 1.309 without improving model-replan success ranking. These results locate useful temporal diagnostics while leaving cheap metrics as the first screening step.**

***Index Terms*—imitation learning, robot learning, checkpoint selection, policy evaluation, action chunking, learned dynamics**

I. Introduction

Training produces many checkpoints: saved versions of a robot policy. Choosing one for deployment differs from reducing its training loss. We study what held-out demonstration metrics reveal about checkpoint quality before execution, and what a learned feedback model adds.

Validation supplies expert observations regardless of the policy's previous errors; execution supplies observations changed by its own actions. A small grasp-position error, for example, can leave the object beside the gripper and alter the next decision. Our motivating manipulator experiments showed different contact behavior among checkpoints with similar logged errors. That experience motivates the question; all reported evidence is simulation-based.

The central finding is a policy-family reversal. Modeled-feedback smoothness improves both ranking and retention on primary Diffusion Policy pools, while cheap metrics lead on BC-RNN. Understanding this contrast requires separating action fit from behavior over time.

We organize the comparisons into three levels: *static checks* measure action fit; *temporal checks* measure action regularity on recorded or predicted observations; *execution-matched diagnostics* compare a chunking policy's old and replacement plans.

PLV examines modeled feedback: the policy proposes an action, dynamics predicts the next observation, and the policy acts on that prediction. The name comes from the proposed latent-validation protocol; in our primary experiments, its state is a normalized vector of robot and object features rather than a learned visual latent. Matched branches isolate the observation source (Fig. 1).

Matched controls distinguish plan disagreement from success prediction.

Our contributions are:

1) We compare three evaluation levels across 1,365 checkpoints, separating full-pool ordering from budgeted candidate retention.
2) We identify a policy-family reversal: modeled-feedback smoothness helps primary Diffusion Policy pools, while cheap metrics lead on BC-RNN.
3) We isolate plan-replacement disagreement with matched observation and sampling controls, and test its selection value across external cohorts.

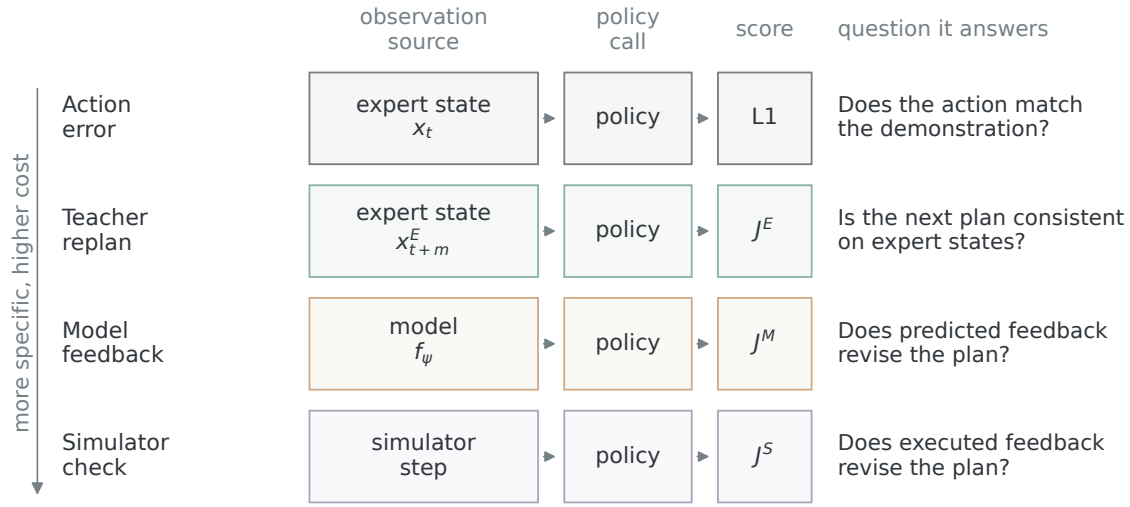


Fig. 1. Action fit and feedback diagnostics. Matched replanning branches share the start, old plan, stride, and sampling seed; only the next observation source changes. Temporal smoothness provides the intermediate comparison (Section IV-B).

II. Related Work

a) Imitation-learning checkpoint selection: Behavior cloning faces covariate shift between expert training states and learner-induced deployment states [1], [2]. Robomimic documents objective–success mismatch [5]; reward-free imitation hyperparameter selection likewise depends on the proxy [13]. We study checkpoint choice within a fixed training run.

b) Offline and model-based policy evaluation: Offline policy evaluation uses logged data to evaluate unexecuted candidates [12], [14], [15]. Decision-targeted models distinguish prediction accuracy from policy-selection accuracy [20]. Visual world-model evaluators estimate robot task success [9], [10], [11]; PLV instead uses compact, task-local models to report behavior diagnostics for many intermediate checkpoints.

c) Failure detection and distribution shift: Runtime monitors and generative-policy uncertainty detect unfamiliar execution states [16], [17], [18]. We test related signals before deployment for checkpoint ranking and rollout triage.

d) Action chunking and replanning stability: Diffusion Policy and ACT predict action *chunks*, partially execute them, then observe and replan [4], [3]. Adaptive and masked execution also study this process [19], [21]. We measure disagreement at the deployed eight-action replacement boundary.

III. What Makes a Useful Checkpoint Metric?

Ranking measures agreement with the full success ordering; *selection* asks whether retaining a fraction of checkpoints preserves a good candidate. Identifying poor early checkpoints can improve ranking without helping distinguish competent later ones.

Let $\Pi = \{\pi_{\theta_1}, \ldots, \pi_{\theta_N}\}$ be checkpoints from one training run and let

$$S_i = \mathbb{E}[\mathbb{1}(\text{task success} \mid \pi_{\theta_i})] \tag{1}$$

denote the unknown deployed success probability. A validation metric produces a score $M_i = M(\pi_{\theta_i}; \mathcal{D}_{\text{val}})$ from held-out demonstrations. We compare its ordering and retained candidates with those obtained from rollout success.

Training-objective validation differs across policies: BC-RNN uses mixture-density likelihood, Diffusion Policy denoising, and ACT chunk reconstruction with latent regularization. Action L1 provides a common action-space comparison against demonstrations.

Spearman correlation measures full-pool ordering; costs are negated so higher positive ρ means better

agreement. Regret@30 asks how much success is lost when retaining the highest-scoring 30%:

$$\text{Regret@}q = S^\star - \max_{i \in \text{Top}_q(M)} S_i, \tag{2}$$

where $q = 0.30$ and $S^\star = \max_i S_i$. Lower is better; zero means a best checkpoint survives. This evaluates retention, not the second-stage choice. Appendices A and I define secondary measures and budgets.

For example, a 30% budget would send 30 of 100 candidates to rollout evaluation, which then chooses among them. Our retrospective curves evaluate first-stage retention using archived outcomes, not the cost of a new online allocation trial.

IV. From Action Error to Replanning Diagnostics

Static and recorded-state temporal checks require policy inference only. PLV adds task-local dynamics; simulator feedback serves as a mechanism check. The comparison changes the observation source while preserving the scored behavior.

A. Level 1: Static action fit

Action L1 asks whether the policy imitates the recorded action:

$$J_{\text{L1}}(i) = \mathbb{E}_{(x_t, a_t) \sim \mathcal{D}_{\text{val}}} \left[\|\pi_i(x_t) - a_t\|_1 \right]. \tag{3}$$

Objective validation loss measures the policy's own training objective, as defined in Section III. Both metrics use recorded observations; neither changes the next input in response to the predicted action.

B. Level 2: Temporal regularity

Under *teacher forcing*, each input comes from the demonstration regardless of earlier predictions. Open-loop smoothness measures action curvature on those inputs:

$$J^E_{\text{smooth}} = \frac{1}{H-2} \sum_{k=2}^{H-1} \|a^E_k - 2a^E_{k-1} + a^E_{k-2}\|_2^2. \tag{4}$$

where $a^E_k = \pi_i(x^E_k)$ and $H \geq 3$ is the number of predicted actions. PLV-1 uses the same curvature statistic, but closes the loop through the dynamics model:

$$\hat{a}_k = \pi_i(\hat{x}_k), \qquad \hat{x}_{k+1} = f_\psi(\hat{x}_k, \hat{a}_k). \tag{5}$$

The suffix "1" denotes a recursive one-step transition model, not the evaluation horizon. Holding curvature fixed isolates modeled feedback.

The main analysis uses normalized low-dimensional observations x_t. Each task-local dynamics ensemble contains five two-layer MLPs of width 256, trained on at most 30,000 demonstration transitions for eight epochs. A secondary audit considers learned representations $z_t = \phi(o_t)$; it establishes no advantage over capacity-matched raw state.

C. Level 3: Execution-matched replanning

A deployed chunking policy commits several actions before observing again. Internal smoothness can hide disagreement with the replacement plan, so we measure boundary changes and revisions to the same future actions.

Diffusion Policy outputs $A_t = \pi(x_t)$ and commits

$$A^{\text{commit}}_t = A_t[s : s+m] \tag{6}$$

before replanning, with offset $s = 1$, stride $m = 8$, prediction horizon 16, two-observation history, and absolute actions.

At replan cycle $r+1$, boundary jump compares the last committed action with the first action in the replacement plan:

$$J_{\text{boundary}} = \frac{1}{d_a} \left\| \frac{A_{t+m}[s] - A_t[s+m-1]}{\sigma_a} \right\|_1. \tag{7}$$

Here σ_a is demonstration action standard deviation. With absolute actions, this measures command discontinuity, not a change between action deltas.

Replan overlap aligns actions that refer to the same future times. If L old-plan actions remain after the executed slice,

$$J_{\text{replan}} = \frac{1}{L d_a} \left\| \frac{B_t - C_{t+m}}{\sigma_a} \right\|_1, \tag{8}$$

where $B_t = A_t[s+m : s+m+L]$ is the retained suffix and $C_{t+m} = A_{t+m}[s : s+L]$ is the aligned replacement prefix. Unlike within-chunk smoothness, this compares two plans for the same future.

a) Separating observation feedback from sampling: Plan changes can reflect new observations or stochastic sampling. We control them separately.

The same-start audit fixes the starting state, old plan, stride, and diffusion seed, changing only the next observation: recorded expert (E), modeled (M), or executed in robosuite (S):

$$A^E_{t+m} = \pi(x^E_{t+m}), \tag{9}$$

$$\hat{x}^M_{t+m} = f^{(m)}_\psi(x_t, A_t), \quad A^M_{t+m} = \pi(\hat{x}^M_{t+m}), \tag{10}$$

$$x^S_{t+m} = \text{Env}^{(m)}(x_t, A_t), \quad A^S_{t+m} = \pi(x^S_{t+m}). \tag{11}$$

Each branch reconstructs the two-observation history from its own source; the simulator branch uses executed states (Appendix F).

Let J^E, J^M, and J^S be branch overlap costs. Subtracting the expert branch measures excess inconsistency:

$$\Delta J^M = J^M - J^E, \qquad \Delta J^S = J^S - J^E. \tag{12}$$

The archive stores costs, not replacement-plan vectors: ΔJ is a scalar proxy, not a direct cross-branch plan distance (Appendix F).

Separate controls change only sampling, only observations with a common seed, or both. Scores average starts within each of four seeds, then take the seed median (Appendix F).

We use *PLV-1* for recursive smoothness at Level 2 and *PLV-M* for model-based replanning at Level 3. The uncertainty-plus-smoothness variant and sampling controls are secondary comparisons; they do not introduce additional evaluation levels.

V. Experimental Protocol

The main retrospective cohort contains 1,365 checkpoints from 45 BC-RNN and Diffusion Policy runs on Can, Lift, Square, Transport, and Tool Hang in robomimic/robosuite [5], [6]. It covers proficient-human (PH), multi-human (MH), machine-generated (MG), and Paired demonstrations where available, with 50 simulator rollouts per checkpoint. This main cohort is retrospective, not preregistered.

The execution archive contains 504 Diffusion Policy checkpoints. Four-seed controls use 130 retained files; same-start uses all 47 locally available nonzero-epoch files in nine outcome-blind grids of five or six checkpoints. These nested mechanism subsets are not complete training grids. External MimicGen, ACT, and RoboCasa cohorts test transfer [7], [8]; Appendix E maps their evidence roles.

Correlations are computed within task–dataset–policy–seed runs, averaged within tasks, then task-balanced. Bootstrap intervals resample tasks and nested runs, not individual checkpoints. Secondary measures include Kendall's τ_b, residual association, and SIMPLER's margin-weighted ordering error (MMRV) [22].

External metrics and inclusion rules were frozen before outcomes; strict prospective tests were preregistered but not fully analyst-blind. The same-start gate was tightened after launch and is post-hoc; inference, dynamics training, simulator stepping, and raw metrics were unchanged.

Rollout noise is assessed by binomial bootstrap, conditional 25/25 splits, and posterior-resolvable pairs, avoiding certain orderings of near-ties (Appendix D).

Ancillary files contain frozen tables, provenance manifests, plotting and audit scripts, and SHA-256 checksums; datasets and policy checkpoints are excluded.

VI. Results

A. RQ1: Can inexpensive metrics retain good checkpoints?

Inexpensive metrics retain strong candidates without a dynamics model. Action L1 and PLV-1 smoothness reach similar full-pool correlations, 0.418 and 0.434, with overlapping intervals (Table I). Retention favors Action L1 (Regret@30 0.033 versus 0.045) and especially open-loop smoothness (0.019). Random retention gives 0.046 with interval $[0.026, 0.064]$; only open-loop smoothness falls below it. The budget curves in Fig. 6 show that this ordering is not confined to a single retained fraction.

Policy family changes the answer. Across 24 Diffusion Policy runs, PLV-1 improves on Action L1 in correlation (0.556 versus 0.462) and Regret@30 (0.006 versus 0.017). Across 21 BC-RNN runs, open-loop smoothness leads. BC-RNN objective loss has slightly lower correlation but better retention than Action L1; in matched external BC-RNN cohorts it leads on both measures. Objective loss is therefore informative within those families even though its pooled result is weak (Appendix B).

Success rises sharply from worst to middle score quintiles but changes little across the top 60%. A metric can separate poor from working policies while having little leverage among the best candidates. This explains why ranking and retention favor different metrics. Finite-rollout noise further limits near-tie comparisons, but does not explain the full gap (Appendices A–D).

B. RQ2: Does modeled feedback add useful information?

Holding the curvature statistic fixed tests whether modeled feedback adds to recorded-state regularity. Open-loop smoothness reaches $\rho = 0.415$, close to PLV-1's 0.434, with better pooled retention. Exact paired tests cannot resolve the small task-level correlation gap.

PLV-1 retains partial rank association 0.183 after Action L1 and open-loop smoothness, and exceeds open-loop correlation by 0.138 on Diffusion Policy. Its value is concentrated in a policy regime, rather than a pooled selection improvement. Recorded-state smoothness is the cheaper first comparison; a modeled loop tests the additional sensitivity to policy-generated observations.

TABLE I

PRIMARY RETROSPECTIVE COHORT: 1,365 CHECKPOINTS, 45 RUNS, FIVE EQUALLY WEIGHTED TASKS. BRACKETS: TASK-CLUSTERED HIERARCHICAL-BOOTSTRAP 95% INTERVALS. ARROWS GIVE PREFERRED DIRECTIONS; RESOLVING ρ USES THE OUTCOME-VARIATION FILTER IN APPENDIX A.

Metric	All-run $\rho \uparrow$ [95% CI]	Regret@30$\downarrow$ [95% CI]	MMRV$\downarrow$	Resolving $\rho \uparrow$
Action L1	0.418 [0.246, 0.579]	0.033 [0.009, 0.069]	0.153	0.334
Open-loop smoothness	0.415 [0.252, 0.569]	0.019 [0.003, 0.049]	0.166	0.353
PLV-1 smoothness	0.434 [0.283, 0.592]	0.045 [0.002, 0.130]	0.160	0.434
PLV-1 uncertainty+smooth	0.425 [0.300, 0.553]	0.041 [0.005, 0.108]	0.161	0.401
Objective validation loss	0.045 [-0.134, 0.231]	0.060 [0.023, 0.105]	0.290	-0.018

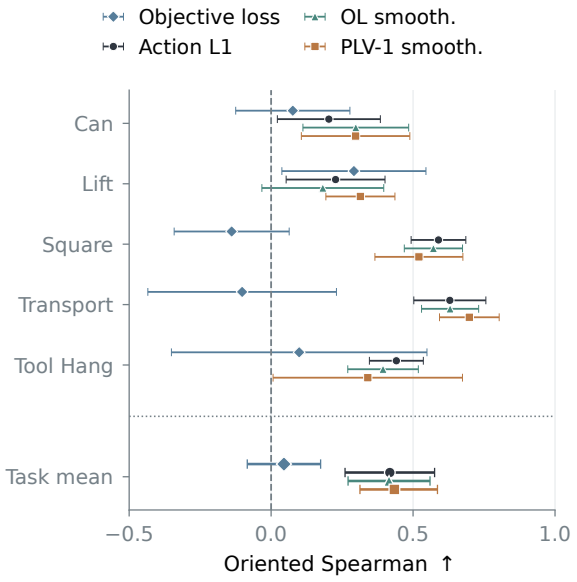


Fig. 2. Task-level correlations from 1,365 checkpoints and 45 runs (±1.96 SE); the summary row uses a task-clustered bootstrap 95% interval. Costs are negated; higher is better. OL denotes open-loop.

C. RQ3: What changes when evaluation matches execution?

We next match the eight-action commitment. This changes the question from whether successive actions are smooth to whether the policy revises actions it had already planned for the same future.

Replacing one-action feedback with eight-action commitment weakens full-chunk smoothness ($\rho = 0.339$). In the hard Diffusion Policy discovery archive, boundary jump and replan overlap correlate more strongly with success (0.730 and 0.707) than full-chunk smoothness (0.339); Action L1 reaches 0.512 (Table II, Panel A; task-pooled discovery estimates).

The retained-checkpoint control tests whether modeled feedback adds to replanning on recorded observations (Table II, Panel B).

For 130 retained checkpoints and four seeds, modeled and recorded feedback differ by task-balanced $\Delta\rho = 0.019$ over 5 tasks (two-sided Wilcoxon $p = 1.000$). Residual association after Action L1 and teacher replan is 0.126 in 9 eligible runs.

Discovery pools checkpoints within two hard tasks; the control compares retained checkpoints within runs and holds sampling fixed. The latter does not reproduce the discovery hard-task ordering. Sampling-only inconsistency also shows that plan changes reflect policy stochasticity. These differences locate a temporal signal, but do not attribute the discovery gain uniquely to modeled feedback.

a) Is prediction error the remaining bottleneck?: Held-out demonstration MSE is 0.0017 at one step and 0.0045 over five steps. On the same 47 checkpoints' policy-generated trajectories, final MSE reaches 3.135 after 16 actions. Doubling model width lowers it to 1.309 without improving success ordering (Appendix H).

Simulator replan inconsistency itself has adverse success ordering ($\rho = -0.212$). Model–simulator agreement is 0.519 for raw costs but only 0.096 for teacher-corrected excess. Thus model error is only one part of the problem: even executed feedback does not turn consistency into a success score. A replacement plan can change because the policy is correcting its behavior, so the magnitude of revision alone does not identify

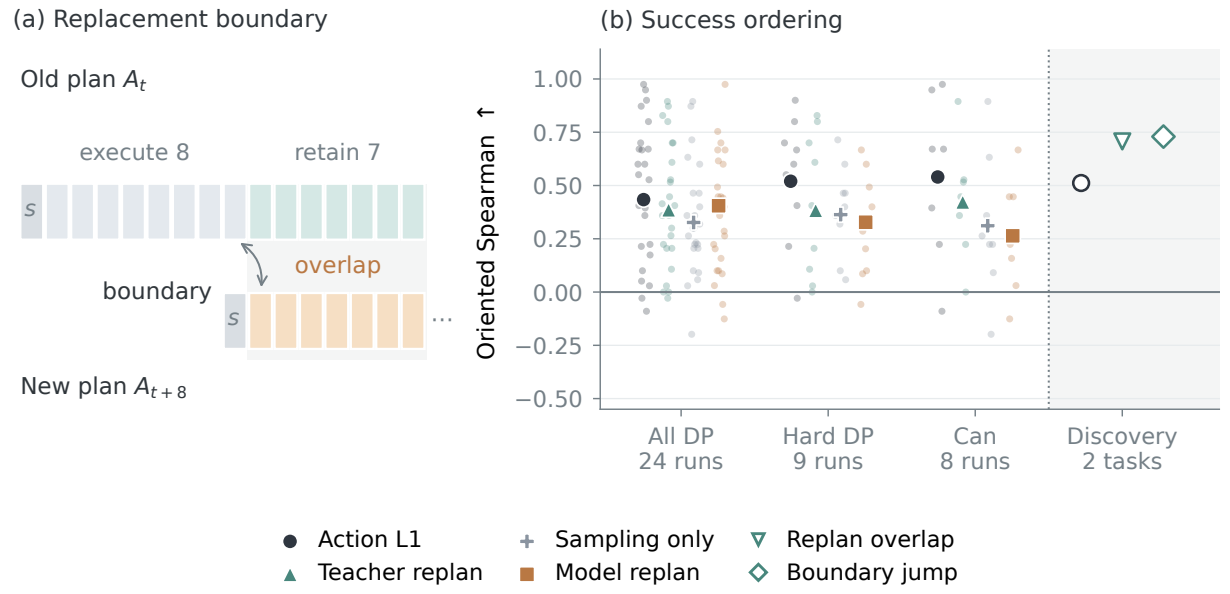


Fig. 3. Plan-replacement diagnostics. (a) Boundary jump and aligned overlap after eight actions, with skip $s = 1$. (b) Filled markers: 130 retained-checkpoint controls (small: runs; large: task-balanced means); hollow: separate task-pooled hard-DP discovery results. Costs are negated.

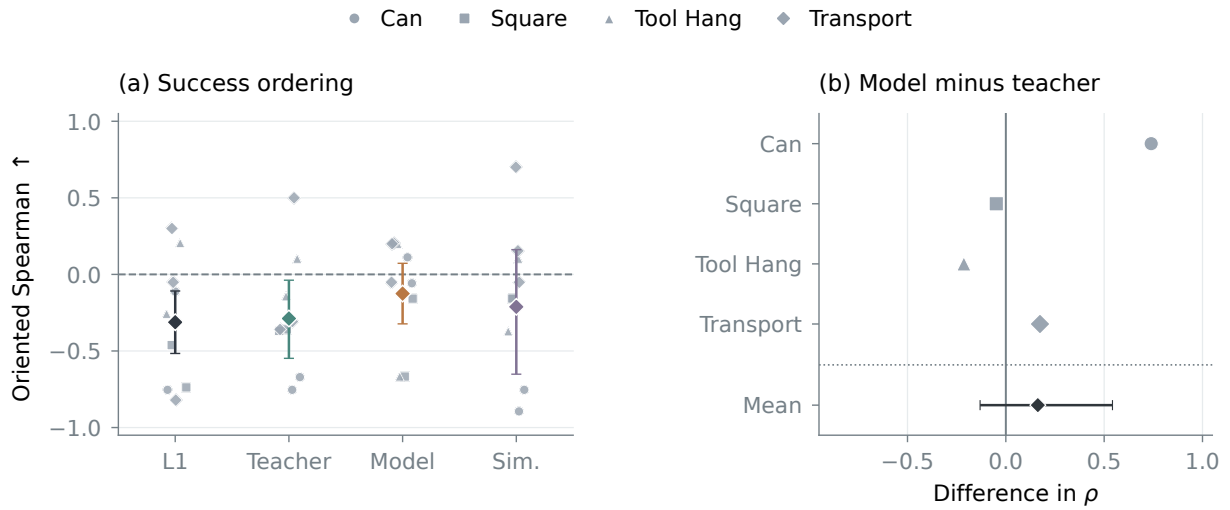


Fig. 4. Same-start controls over 47 checkpoints and nine runs. (a) Oriented success correlations: runs (gray), tasks (shapes), and task-balanced means (colored diamonds; task-clustered 95% intervals). Costs are negated. (b) Model-minus-teacher correlation by task and its task-balanced mean.

whether that change is helpful.

Teacher-corrected model excess fails the post-hoc acceptance criterion: success correlation 0.068, residual −0.309 after Action L1 and teacher replan in six eligible runs. Its descriptive regret gain (0.017 versus 0.031) comes from two improved tasks and two ties (exact

TABLE II

EXECUTION-MATCHED REPLANNING. A: TWO TASK-POOLED HARD-DP DISCOVERY ESTIMATES. B: 24 RETAINED-CHECKPOINT RUNS (130 FILES); RESIDUAL ρ CONTROLS ACTION L1 AND TEACHER REPLAN IN 9 ELIGIBLE RUNS. PANELS USE DIFFERENT AGGREGATION UNITS AND ARE NOT POOLED. COSTS ARE NEGATED; DASHES DENOTE UNDEFINED QUANTITIES.

Metric	All DP $\rho \uparrow$	Hard DP $\rho \uparrow$	Can control $\rho \uparrow$	Residual $\rho \uparrow$
A. Historical hard-DP discovery (2 task-pooled estimates)				
Action L1	–	0.512	–	–
One-step PLV smoothness	–	0.745	–	–
Chunk PLV smoothness	–	0.339	–	–
Boundary jump	–	0.730	–	–
Replan overlap	–	0.707	–	–
B. Four-seed retained-binary control (24 task/dataset/seed runs)				
Four-seed Action L1	0.434	0.520	0.540	–
Teacher-forced replan	0.386	0.384	0.423	–
Sampling-only inconsistency	0.327	0.363	0.311	–
Model-induced replan	0.405	0.328	0.264	0.126
Model+sampling replan	0.337	0.316	0.313	0.066

TABLE III

SAME-START FEEDBACK: RECORDED (E), MODELED (M), EXECUTED (S). NINE OUTCOME-BLIND RETAINED GRIDS, 47 NONZERO-EPOCH CHECKPOINTS, FOUR TASKS; SHARED STARTS, STRIDE, AND FOUR SEEDS. ΔJ SUBTRACTS EXPERT-STATE COST, NOT PLAN VECTORS. COSTS ARE NEGATED; AGREEMENT CORRELATES ΔJ^M WITH ΔJ^S.

A. Success ordering

Task	Runs	Action L1	J^E	J^M	J^S
Can	2	-0.433	-0.712	0.027	-0.824
Square	2	-0.600	-0.364	-0.412	-0.156
Tool Hang	2	-0.026	-0.020	-0.233	-0.134
Transport	3	-0.191	-0.056	0.118	0.268
Task-balanced	9	-0.312	-0.288	-0.125	-0.212

B. Teacher-corrected ordering

Task	Runs	ΔJ^M	ΔJ^S	Agree. ρ	Residual ρ
Can	2	0.712	-0.499	-0.221	–
Square	2	-0.469	-0.053	0.300	-0.399
Tool Hang	2	0.006	-0.405	0.071	-0.095
Transport	3	0.021	0.274	0.233	-0.432
Task-balanced	9	0.068	-0.171	0.096	-0.309

$p = 0.500$). Here the useful role is to identify disagreement for rollout investigation, rather than automatically rerank checkpoints.

The model-minus-teacher correlation difference

TABLE IV

NOISE AND DYNAMICS CHECKS. FIRST THREE ROWS: PRIMARY COHORT (45 RUNS, 1,365 CHECKPOINTS); DYNAMICS ROWS: SAME-START AUDIT (NINE RUNS, 47 CHECKPOINTS). RELIABILITY CONDITIONS ON 50 OUTCOMES. NORMALIZED-STATE MSE COMPARES SEPARATE DEMONSTRATION (1/5-STEP) AND POLICY-INDUCED (16-STEP ENDPOINT) PROTOCOLS.

Check	Result Reading
25/25 split $\rho \uparrow$	0.500 [0.472, 0.529] substantial noise
Full-50 reliability$\uparrow$	0.667 noise is not the sole limit
Random Regret@30$\downarrow$	0.046 Action L1 0.033, PLV-1 0.045
Demo dynamics MSE$\downarrow$	0.0017 / 0.0045 1 / 5 steps
Same-start final MSE$\downarrow$	3.135 16-step drift
Simulator J^S $\rho \uparrow$	-0.212 adverse ordering

was 0.163 (task-clustered bootstrap 95% interval $[-0.131, 0.542]$; Fig. 4), which spans zero.

D. RQ4: Does the result transfer to new tasks and policies?

External comparisons reinforce the distinction between a policy-family result and a universal selector. Open-loop smoothness leads the broader action-chunking cohort ($\rho = 0.829$). Selected MimicGen candidates are positive, while complete BC-RNN grids, unseen ACT, and six strict RoboCasa cohorts do not establish a general PLV gain (Appendix E).

Two preregistered cohorts each froze 126 scores before 6,300 rollout episodes. Both failed acceptance: Regret@30 worsened from 0.007 to 0.017 on domain variants and 0.047 to 0.050 on an unseen bimanual family. A candidate-only cohort yielded three successes in 2,000 episodes, too little variation to compare selectors.

These tests evaluate particular score variants, not execution-matched overlap on every family. Together they show why the appropriate baseline and the outcome spread must be checked before carrying a screening rule to a new setting.

VII. IMPLICATIONS FOR SCREENING

The comparison gives a practical order of questions: action fit first, temporal regularity next, then plan revision for chunking policies. Open-loop smoothness supports inexpensive retention overall; PLV-1 adds value on the primary Diffusion Policy subset. Replanning characterizes feedback-sensitive behavior among surviving candidates. When offline evidence conflicts, rollout can settle the choice.

This assigns dynamics a specific diagnostic job instead of using it for every ranking decision. To measure end-to-end savings, a prospective allocation trial would freeze the screen and compare selected success at equal episode budgets with cheap-only and random allocation. Offline fitting and inference costs belong in that comparison. The existing frozen trial had too little outcome variation to resolve it; prospective savings remain unestablished.

VIII. DISCUSSION AND LIMITATIONS

a) Policy family is part of the evaluation problem: A pooled table hides the policy-family reversal: cheap checks describe BC-RNN well, while primary Diffusion Policy pools benefit more from modeled temporal regularity. The hard-task discovery result further points to the plan-replacement boundary, rather than smoothness throughout the chunk.

b) Why ranking and selection disagree: Full-pool ranking rewards separating poor checkpoints from working ones. Selection concentrates on the much smaller differences among survivors. Among the retained top 30%, estimated success-rate spread after rollout-noise correction is only 0.042, with more room on Tool Hang (0.083) than Can (0.008). The opportunity for an additional diagnostic is therefore task-dependent; a better overall correlation need not improve the deployment choice.

c) Prediction quality is not decision quality: Better prediction need not improve selection:

$$\text{transition calibration} \not\Rightarrow \text{checkpoint ordering}. \tag{13}$$

A probabilistic ensemble improves uncertainty–error calibration without a comparable ranking gain. Aleatoric transition noise may predict local error without distinguishing checkpoints; this remains a possible explanation. Together with the simulator control, the result points to the scoring target as well as dynamics fidelity.

Evaluation must capture whether a plan revision helps the task, not merely whether the next state or the revision is predictable.

d) Limitations: The primary comparison has complete archived outcomes, whereas mechanism audits use retained checkpoint files. Differences in policy family, recurrence, stride, and task difficulty are not independently varied, so they cannot establish action chunking as the cause of the reversal. Model results cover five-member MLP ensembles, 30,000 transitions, eight epochs, and widths 256/512, not a capacity-scaling law.

Low-dimensional simulation does not validate real-robot transfer. A small real-manipulator study could test whether the family-specific ordering and plan-replacement behavior survive perception and physical execution. Together with an informative prospective budget trial, this would connect the current screening evidence to deployment cost.

Strict external tests were preregistered but not fully analyst-blind: integrity checks exposed filename scores and partial rewards, not episode labels or confirmatory reports; analysis remained unchanged. Same-start acceptance is post-hoc. Conditional 25/25 reliability omits unarchived initial-condition and environment-seed variation (Appendices D, E, and G).

IX. Conclusion

Across 1,365 checkpoints, open-loop smoothness gave the best pooled Regret@30, while modeled feedback improved ranking and retention on the primary Diffusion Policy pools. Plan disagreement marked hard-task behavior, but its discovery advantage did not persist in matched retained-checkpoint controls. These results favor inexpensive shortlisting followed by replanning diagnostics where temporal behavior matters. Whether this workflow saves rollout episodes remains to be tested in a prospective allocation trial.

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Appendix A
Frozen Analysis Contract

The analysis unit is a task–dataset–policy–seed training run. Checkpoint correlations are computed within runs, averaged over runs within each task, then averaged with equal task weight. Task-clustered bootstrap intervals resample tasks and nested runs. Outcome-resolving runs require success IQR at least 0.10, at least five distinct success values, and a best–worst success gap at least 0.20.

Regret@q retains $\lceil qN \rceil$ checkpoints; Regret@1 retains exactly one. Lower-cost metrics sort first; boundary ties use checkpoint path in the frozen manifest, never rollout success. Each primary checkpoint has 50 simulator rollouts (0.02 success increments).

SIMPLER's MMRV averages each checkpoint's largest true-success margin to a checkpoint incorrectly ordered by the metric, estimating margins from rollout rates. Lower is better; large reversals count more than unresolved near-ties. Primary task-balanced values are 0.153 for Action L1, 0.160 for PLV-1 smoothness, 0.166 for open-loop smoothness, and 0.290 for objective loss.

Appendix B
Policy-Family Split and Design Resolution

With five task means, paired differences detectable at 80% power are: PLV-1 smoothness versus Action L1 (observed $\Delta\rho = 0.016$, detectable $\Delta\rho \approx 0.117$); PLV-1 smoothness versus open-loop smoothness (observed $\Delta\rho = 0.019$, detectable $\Delta\rho \approx 0.101$); open-loop smoothness versus Action L1 (observed $\Delta\rho = -0.003$, detectable $\Delta\rho \approx 0.073$). No observed gap reaches 0.19 of its detectable difference. Comparisons are unresolved, not proven equivalent; the inference unit is the task, not the checkpoint.

All metric–success relations show curvature: a quadratic term raises R^2 by 0.15–0.19 over a linear fit, with consistently negative quadratic coefficient; LOWESS averages 1.4–1.7 turning points per run. We therefore summarize by ranks; linear fits overstate late-training gains.

Action-space and PLV smoothness signals have mean within-run pairwise rank correlations 0.53–0.66; the first rank component explains 73.9% of variance. PLV-1's partial rank association after Action L1 and open-loop smoothness is 0.183: remaining association, not an established selection gain.

Appendix C
Score Normalization Sensitivity

Normalization is pool-dependent, not intrinsically calibrated. Task-balanced Spearman is 0.366 with target-pool median/IQR, 0.364 with a fixed seed reference, 0.321 with within-run empirical ranks, and 0.178 without normalization. Random half-pools retain ordering ($\rho = 0.961$) but shift scores by 0.186 on average. Score magnitudes are interpreted only within the declared pool.

Appendix D
Finite-Rollout Reliability and Baselines

A binomial bootstrap resamples outcomes at each checkpoint's observed rate, preserving the run/task hierarchy.

The primary archive stores success counts, not episode order. Conditional on each 50-episode count, we draw one 25-episode half from the exact hypergeometric distribution and assign the remaining successes to the other. Over 1,000 repetitions, task-balanced split-half Spearman is 0.500 with interval $[0.472, 0.529]$; Spearman–Brown reliability for 50 outcomes is 0.667. This diagnoses conditional reliability, not correlation with an unobserved true success probability.

Checkpoint i with k_i successes receives a Beta$(k_i + 1, 50 - k_i + 1)$ posterior. Pair ordering is scored only when $P(S_i > S_j) > 0.95$ or its reverse exceeds 0.95; adjacent observed rates such as 0.48 and 0.50 do not establish certain performance differences.

At 30%, random mean regret is 0.046 with randomization interval $[0.026, 0.064]$; selecting the last checkpoint gives 0.118. Table IX reports other baselines. With only five nonzero task differences, the smallest attainable exact two-sided signed-rank p-value is 0.0625.

Appendix E
Cohorts and Evidence Roles

a) External blinding deviation: Strict prospective external evaluations are preregistered, not fully analyst-blind. Methods, budgets, gates, confirmatory analysis, and reporting code were frozen before target outcomes. A filesystem inventory printed validation scores embedded in checkpoint filenames; a live integrity audit exposed normalized partial sparse rewards for one task before aggregate ranking analysis. No episode-level outcomes, registered confirmatory decisions, all-budget endpoint reports, or paired bootstrap results were exposed. The selector script and thresholds were already fixed; registered analysis remained unchanged.

TABLE V

PRIMARY POLICY-FAMILY SPLIT: PLV-1 LEADS ON DIFFUSION POLICY, OPEN-LOOP SMOOTHNESS ON BC-RNN. OBJECTIVE LOSS IS ADVERSE ON DP BUT COMPETITIVE ON BC-RNN. DESCRIPTIVE FIVE-TASK SUBGROUP ESTIMATES; NO MULTIPLICITY CORRECTION. ARROWS GIVE PREFERRED DIRECTIONS.

Metric	$\rho \uparrow$	MMRV$\downarrow$	Regret@30$\downarrow$
BC-RNN (21 runs)			
Action L1	0.372	0.190	0.050
Open-loop smoothness	0.401	0.184	0.028
PLV-1 smoothness	0.303	0.231	0.084
PLV-1 uncertainty+smooth	0.344	0.224	0.076
Objective validation loss	0.360	0.261	0.034
Diffusion Policy (24 runs)			
Action L1	0.462	0.120	0.017
Open-loop smoothness	0.418	0.149	0.010
PLV-1 smoothness	0.556	0.093	0.006
PLV-1 uncertainty+smooth	0.502	0.102	0.008
Objective validation loss	-0.243	0.316	0.083

TABLE VI

COHORT MAP. COUNTS ARE NON-ADDITIVE: THE 504, 130, AND 47 ROWS ARE NESTED PRIMARY-POOL SUBSETS. SELECTED GRIDS ARE NOT COMPLETE-GRID CONFIRMATIONS.

Cohort	Policy	Tasks/ runs	Ckpts	Rollouts/ ckpt	Grid	Evidence role
Primary retrospective	BC-RNN, DP	5/45	1,365	50	heterogeneous	main analysis
Execution archive	DP	5/24	504	50	archived	development
Four-seed controls	DP	5/24	130	50	retained	post-hoc control
Same-start branches	DP	4/9	47	50	retained	mechanism
Coffee/StackThree	BC-RNN	2/8	328	50	complete	frozen
Eight-task breadth	BC-RNN	8/32	1,312	50	complete	frozen
Action-chunk breadth	DP	8/8	168	50	complete	frozen
Unseen family	ACT	2/6	126	50	complete	transfer
Domain variants	ACT	2/6	126	50	complete	strict prospective
Unseen semantic family	ACT	2/6	126	50	complete	strict prospective
Strict sequence	ACT	17/51	561	50	complete	prospective
Candidate-only stress	ACT	3/9	40	50	selected	non-discriminative

Filename scores and intermediate rewards were not fully embargoed.

APPENDIX F
DIFFUSION POLICY EXECUTION CONTRACT

The low-dimensional Diffusion Policy uses prediction horizon 16, observation horizon two, executable offset $s = 1$, stride $m = 8$, and absolute actions. The committed slice is $A_t[s : s + m]$. Action L1 compares the sampled executable slice with the aligned expert slice. The global metric uses one sampled evaluation; execution controls use four branch-matched diffusion seeds and take their median after averaging validation starts.

Teacher, model, and simulator branches share start,

TABLE VII

EXTERNAL EVIDENCE. "CONFIRM." REQUIRES FROZEN, COMPLETE, OUTCOME-RESOLVING COVERAGE; "DESCRIPTIVE" MEANS SELECTED OR INCOMPLETE COVERAGE; "FAILED" MEANS AN UNCONFIRMED FROZEN PLV CLAIM; "N/D" MEANS NON-DISCRIMINATIVE OUTCOMES.

Cohort	Policy	Com-plete	Frozen	Resol-ving	Action/objective evidence	PLV evidence
Coffee/StackThree	BC-RNN	Yes	Yes	Yes	objective strongest	failed vs. baseline
Eight-task breadth	BC-RNN	Yes	Yes	Yes	objective strongest	failed vs. baseline
Action-chunk breadth	DP	Yes	Yes	Yes	open-loop strongest	below Action L1; gate not met
MimicGen selected	DP	No	Yes	Yes	not tested	descriptive
Retained recurrent	BC-RNN	No	Yes	Yes	not tested	descriptive
Unseen family	ACT	Yes	Yes	Yes	action signal	failed confirmation
Domain variants	ACT	Yes	Yes	Yes	not tested	failed gate
Unseen semantic family	ACT	Yes	Yes	Yes	not tested	failed gate
Strict RoboCasa	ACT	Yes	Yes	Mixed	not promoted	failed gates
Candidate-only	ACT	No	Yes	No	N/D	N/D

old plan, stride, and sampling seed. At the next replan, the teacher reads demonstration history; the model shifts history by one observation per predicted transition. The simulator restores the demonstration simulator-state snapshot, executes the same committed absolute-action slice converted to the environment convention, and appends resulting observations.

With four policy seeds and distance d between aligned plans, we separate diffusion stochasticity:

$$J_{\text{sample}} = d(\pi(x;\xi_1), \pi(x;\xi_2)), \tag{14}$$

$$J_{\text{state}} = d(\pi(x;\xi), \pi(x';\xi)), \tag{15}$$

$$J_{\text{total}} = d(\pi(x;\xi_1), \pi(x';\xi_2)). \tag{16}$$

These change sampling only, state only (common random numbers), or both (stochastic deployment), respectively. Common random numbers reuse the diffusion sampling seed. High sampling-only distance indicates policy stochasticity, not closed-loop state sensitivity.

The clean mechanism contrasts are

$$D_{M,E} = d(A^M, A^E), \qquad D_{S,E} = d(A^S, A^E), \tag{17}$$

$$D_{M,S} = d(A^M, A^S). \tag{18}$$

The completed 47-checkpoint archive stored only branch distances to the old-plan suffix, supporting the weaker excess proxy $\Delta J^M = J^M - J^E$, not reconstruction of direct contrasts. The evaluator now records direct common-seed plan distances; these are not completed manuscript results.

APPENDIX G
SAME-START AVAILABILITY AND DEVIATION LOG

The audit includes all 47 locally available nonzero-epoch binaries in nine retained task–dataset–seed grids: 6,016 observations (47×32 starts $\times 4$ policy seeds), no duplicate checkpoint/start/seed keys, and success-independent inclusion. Runs contain five or six checkpoints; conditional residuals are finite in six of nine runs.

Launch configuration was reconstructed during execution from the running job's launch arguments and the fixed experiment queue. After launch, the aggregation gate was tightened to at least eight finite runs. Policy inference, dynamics training, simulator stepping, and raw row generation were unchanged. The final gate is therefore labeled post-hoc rather than preregistered.

Two raw fidelity/ordering gates pass; the teacher-corrected gate fails. Task-balanced excess-success correlation is 0.068, model–simulator excess agreement 0.096, and residual association after Action L1 and teacher replan −0.309 over six eligible runs. Descriptive Regret@30 is 0.017 for the excess proxy versus 0.031 for Action L1. Only two task differences are nonzero (exact two-sided Wilcoxon $p = 0.500$), insufficient for automatic reranking.

Task-balanced normalized-state MSE on held-out demonstrations is 0.0017 at one step and 0.0045 averaged over five demonstration steps. In the same-start audit, one-step policy-induced MSE is 0.063; final imagined–simulator MSE after 16 actions is 3.135. Raw

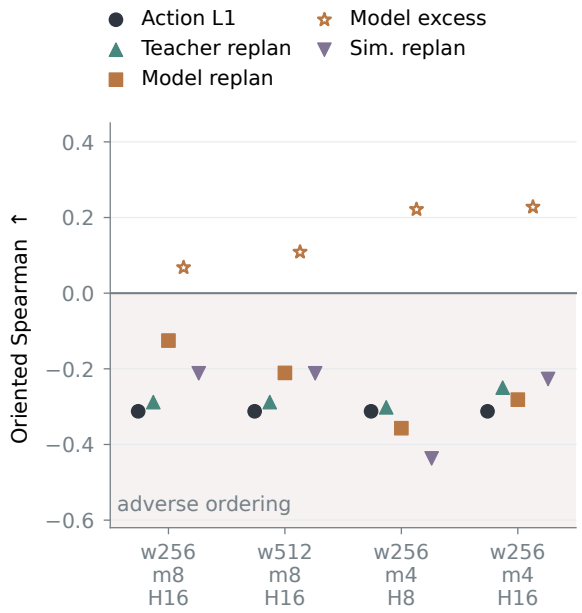


Fig. 5. Capacity and stride do not repair model-replan ordering. Points are task-balanced over nine same-start runs (47 checkpoints, four tasks). Negated costs make higher correlation better; shading marks adverse ordering. All replan branches remain nonpositive. Hollow stars denote derived teacher-corrected excess proxies, not measured branch scores. Evidence grade: mechanism audit.

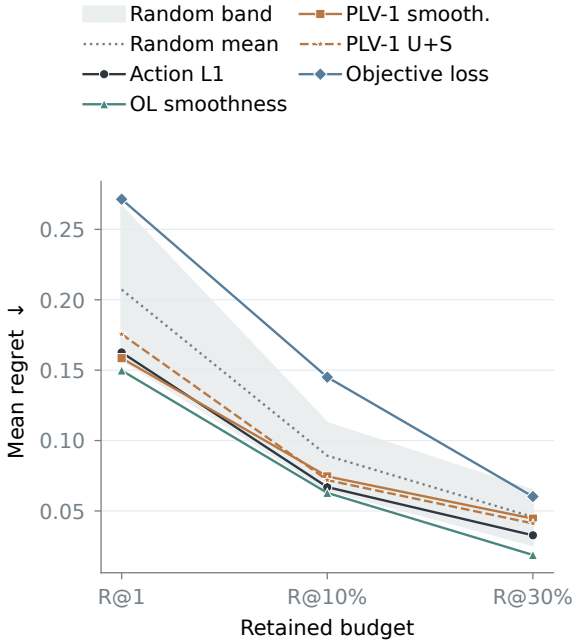


Fig. 6. Budget sweep over 45 runs and five tasks; lower regret is better. The band shows the task-balanced random-selector 95% interval (10,000 draws), the dotted line its mean. Only open-loop smoothness is below the band at all three budgets. Action L1 and PLV-1 track closely; objective loss is above the band at R@1/R@10% and inside at R@30%. Evidence grade: primary.

model–simulator replan ordering is 0.519, versus 0.096 for teacher-corrected excess. Simulator-induced replan has task-balanced $\rho = -0.212$: learned-model error is not the only bottleneck.

The 504-checkpoint audit does not support sparse failures being hidden by averaging. Direct mean replan has $\rho = 0.557$, versus p90 0.448 and worst-20% CVaR 0.418; recursive values are 0.551, 0.498, and 0.459.

Appendix H
Controlled Stride and Capacity Check

Every condition in the controlled stride and capacity audit uses the same 47 retained checkpoints, 32 validation starts from the common 16-action candidate set, and four common diffusion seeds. Valid replanning contrasts are stride four versus eight at horizon 16 and horizon eight versus 16 within stride four. Stride eight, horizon eight has one policy call and is a fidelity-only control.

At deployed stride and horizon, width 512 changes oriented model-replan rho by −0.085 versus width 256; only two of nine paired runs improve. The task-bootstrap interval is $[-0.245, 0.142]$, exact paired Wilcoxon $p = 0.625$ (minimum 0.125 for four nonzero task means). Lower MSE establishes no ranking gain under the same eight-epoch budget.

Appendix I
Budget Sensitivity and Outcome-Noise Scale

Table IX uses task-balanced aggregation and 10,000-draw randomization at top one and 10%/30% retained budgets. Top one chooses a single checkpoint; the other budgets retain candidate sets. PLV-1 has lower regret than Action L1 only at top one and is never best. At 30%, both lie inside the random interval; open-loop smoothness is below it.

With conditional split-half Spearman $r_{25} = 0.500$, Spearman–Brown reliability is $r_{50} = 2r_{25}/(1 + r_{25}) = 0.667$. Assuming noiseless metrics gives the one-sided classical attenuation approximation $\rho_{\text{true}} \approx$

TABLE VIII

CONTROLLED SENSITIVITY, TASK-BALANCED OVER NINE RETAINED RUNS. NEGATED REPLAN COSTS MAKE HIGHER ρ BETTER; MODEL-REPLAN CORRELATION AND FIDELITY MSE APPEAR ONLY WHERE DEFINED.

Condition	Replan ρ	Final MSE	Reading
m=8, H=8, w=256	–	0.408	fidelity only
m=4, H=8, w=256	-0.357	0.557	shorter horizon
m=4, H=16, w=256	-0.281	3.270	stride check
m=8, H=16, w=256	-0.125	3.135	baseline
m=8, H=16, w=512	-0.211	1.309	lower MSE, worse rank

TABLE IX

TASK-BALANCED REGRET (LOWER IS BETTER). BRACKETS GIVE THE RANDOM-SELECTOR 95% INTERVAL SHARED WITHIN EACH BUDGET.

Selector	R@1	R@10%	R@30%
Random (mean)	.207	.089	.046
Random (95% interval)	[.158, .266]	[.065, .112]	[.026, .064]
Action L1	0.163	0.067	0.033
PLV-1 smoothness	0.159	0.075	0.045
Open-loop smoothness	0.150	0.063	0.019

$\rho_{\text{observed}}/\sqrt{r_{50}}$. Action-L1, PLV-1, and objective-loss values become approximately 0.512, 0.532, and 0.055. This is neither a rank-specific correction nor a true-performance ceiling: the split conditions on archived success counts, omitting unarchived initial-condition or environment-seed variation.

APPENDIX J
INTERPRETING SCREENING RESULTS BY REGIME

These regime interpretations are not deployment rules: the screening cascade still requires an informative frozen prospective allocation trial.

a) Strict prospective chronology: Each of two cohorts registered datasets, source-model hashes, environment, target tasks, and gate before target training, computed outcome-blind metrics for 126 checkpoints, and froze scores before 6,300 binary rollout episodes. No target outcome was used to fit, select, or recalibrate the scorer. Mean Spearman moves from 0.681 to 0.629 on the domain-variant cohort (Regret@20 0.010 → 0.017, Regret@30 0.007 → 0.017, 4/6 pools stable) and from 0.304 to 0.279 on the unseen bimanual dexterous family (Regret@20 0.050 → 0.050, Regret@30 0.047 → 0.050, 5/6 stable). Both fail their complete preregistered gates. The semantic-family cohort was the registered decisive generalization test; the domain-variant cohort is complementary, not a replacement.

APPENDIX K
WHERE THE DISCRIMINATIVE SIGNAL SITS

Across five metrics, worst-to-middle quintile gains span 0.133–0.273 in task-balanced success; the top 60% spans only −0.062–0.026. Top-quintile resolution, unlike global ρ, follows Regret@30 ordering (descriptive, not a test). The run-best success still exceeds each metric's best-quintile mean by about 0.14.

Within runs, action-space and smoothness scores track epoch at ρ = 0.56–0.69, success at 0.47. Yet selecting the last checkpoint performs poorly: success plateaus then declines late. Cheap metrics help avoid that decline more than rank survivors. Objective loss instead correlates with epoch at −0.353, tracking objective overfitting decoupled from deployed success and adverse on Diffusion Policy.

APPENDIX L
HOW MUCH IS LEFT TO WIN

Binomial decomposition estimates underlying SD 0.042 and headroom 0.063, versus observed gap 0.110. The 50-episode winner matches the underlying best with

TABLE X

INTERPRETATION BY REGIME, NOT A PROSPECTIVELY VALIDATED CASCADE. LATER STAGES DO NOT AUTOMATICALLY REPLACE EARLIER RANKINGS.

Regime	Primary signal	Decision
Heterogeneous pool	Action L1	default filtering
Matched BC-RNN	objective loss + Action L1	validate locally
DP near-tie (unvalidated)	teacher-forced replan	flag temporal disagreement
Hard/contact DP	boundary + overlap	discovery-only flag
Material disagreement	real rollout	abstain from reranking
New family / flat outcomes	recalibrate	no transfer claim

TABLE XI

PRIMARY TASK-BALANCED SUCCESS BY WITHIN-RUN SCORE QUINTILE (Q1 WORST, Q5 BEST; Q2/Q4 OMITTED). Q1–Q3 IMPROVEMENT IS LARGER THAN Q3–Q5 IMPROVEMENT, DISTINGUISHING FULL-POOL RANKING FROM TOP-CANDIDATE CHOICE.

Metric	Q1	Q3	Q5	Q1→Q3	Q3→Q5
Open-loop smoothness	0.277	0.509	0.535	+0.232	+0.026
Action L1	0.252	0.517	0.531	+0.265	+0.014
PLV-1 smoothness	0.254	0.524	0.532	+0.270	+0.008
PLV-1 uncertainty+smooth	0.257	0.531	0.524	+0.273	-0.006
Objective validation loss	0.375	0.508	0.445	+0.133	-0.062

TABLE XII

ABILITY-SPREAD ESTIMATE WITHIN ACTION L1'S RETAINED TOP 30%. OBSERVED GAP: SAMPLED MAXIMUM MINUS MEAN. BINOMIAL DECOMPOSITION ESTIMATES UNDERLYING SD AND BEST-MINUS-MEAN HEADROOM; HIT RATE ESTIMATES HOW OFTEN THE 50-EPISODE WINNER IS THE UNDERLYING BEST. PRIMARY RETROSPECTIVE EVIDENCE.

Task	Observed gap	True SD	Headroom	Hit rate
Can	0.046	0.008	0.013	0.162
Lift	0.090	0.049	0.070	0.420
Square	0.077	0.011	0.018	0.160
Transport	0.141	0.062	0.092	0.366
Tool Hang	0.198	0.083	0.120	0.501
Task-balanced	0.110	0.042	0.063	0.322

probability 0.322; near-ties limit expected selection loss to 0.012. Estimated SD is 0.072 on Transport/Tool Hang versus 0.009 on Can/Square.

Regret@30 uses a 50-episode maximum, mixing ability and sampling noise: its absolute scale is not all recoverable performance. Retained Can and Square candidates are nearly interchangeable, leaving little selection headroom; Transport and Tool Hang offer more, where execution-matched discovery was strongest.

APPENDIX M
EVIDENCE GRADES

a) Confirmatory support: Metric and endpoint were frozen before outcomes, the grid is complete, the cohort is outcome-resolving, and the declared comparison is met.

b) Descriptive support: Positive results use selected, incomplete, availability-limited, or metric-development coverage.

c) Failed confirmation: A frozen claim or promotion gate fails on complete, outcome-resolving coverage.

d) Non-discriminative: The prespecified outcome-variation criterion fails; neither positive nor negative selector evidence is assigned.

Selected MimicGen and retained recurrent results are descriptive. Coffee/StackThree, eight-task BC-RNN, action-chunk breadth, and unseen ACT use complete grids. Strict RoboCasa has frozen promotion gates but mixed outcome resolution; candidate-only RoboCasa is non-discriminative.

Appendix N
Artifact Map

The submission's `anc/results/` contains frozen manifests, run/pair summaries, bootstrap results, raw same-start rows, score-resolution and fidelity-reuse Parquet tables, gates, and historical checksums; `anc/code/` supplies selected generation, audit, and evaluator scripts. Each new Parquet table retains adjacent metadata JSON with input/output hashes, random seed, and Git commit. `anc/README.md` describes the inventory; `anc/SHA256SUMS` verifies submission bytes. Historical workspace paths require remapping; this is not a complete training repository. The cost–regret plot is omitted because runtimes lack a common hardware, batching, and reset benchmark.